# Strategic Analysis Report: Hotel Booking Demand & Revenue Optimization

**Date:** March 2026

**Tools Used:** SQL (SQLite), VS Code, DB Browser

**Dataset:** Kaggle Hotel Booking Demand (119,390 records)

## 1. Executive Summary

This report provides a multi-dimensional diagnostic of the hotel group's operational health. Our analysis reveals a significant **bifurcation** between City and Resort hotel performance. While **City Hotels** drive high volume and premium last-minute pricing, they suffer from **a critical 42% cancellation rate**. Conversely, **Resort Hotels** maintain higher stability but struggle with "perishable inventory" price slashing for late bookings.

**Key Strategic Pivot:** Shift focus from "Booking Volume" to "Booking Quality" by tightening cancellation policies for long-lead orders and aggressively targeting high-yield Western European transient markets.

## 2. Phase 1: Data Integrity & Sanitization

Before analysis, the dataset was sanitized to ensure business logic consistency:

**Ghost Bookings:** Removed records with 0 adults, children, and babies.

**Price Anomalies:** Filtered out negative ADR (Average Daily Rate) values.

**Missing Values:** Categorized undefined countries as "Unknown" and null agents/companies as "0".

*Final cleaned dataset: ~ 117,399 valid records.*

## 3. Phase 2: Global Overview (The Landscape)

Comparing the two properties reveals a fundamental difference in risk profiles:

| Hotel | Total bookings | Cancel rate | Avg_adr | Avg_lead_time | Avg_stay_nights |
|---|---|---|---|---|---|
| City Hotel | 78,092 | 42.22 | 106.95 | 111.0 | 3.0 |
| Resort Hotel | 39,307 | 28.08 | 96.77 | 93.4 | 4.37 |

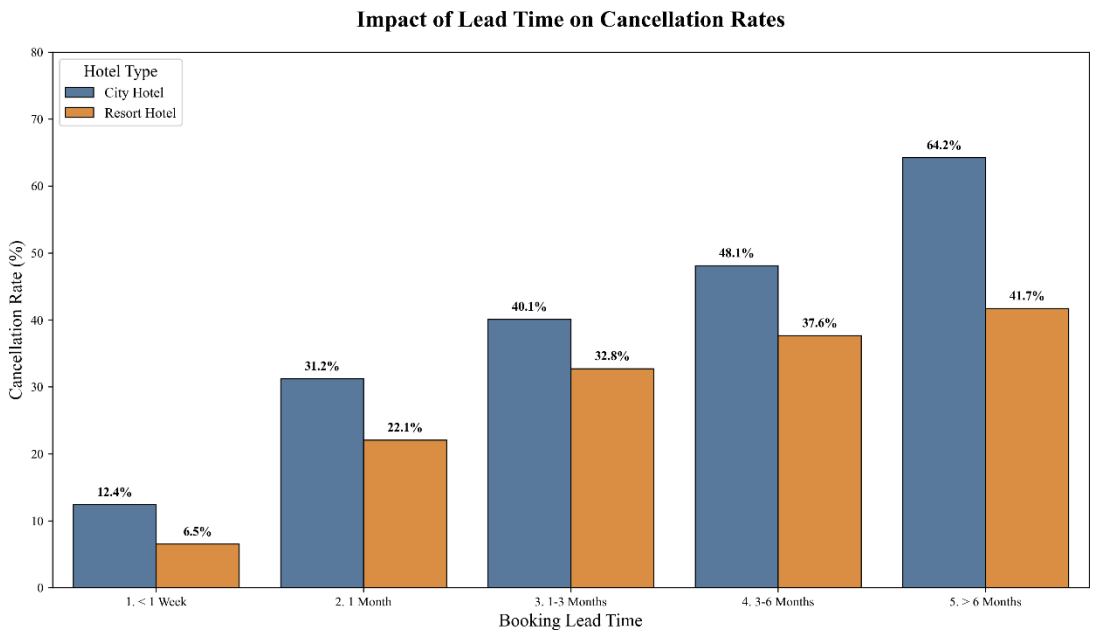
Compared with Resort Hotel, City Hotel has a significantly higher cancel rate despite its higher number of total bookings.

## 4. Phase 3: Loss Diagnosis (The Cancellation Crisis)

### 4.1 The "Death Zone" of Lead Time

We discovered that lead time is the strongest predictor of cancellation. For City Hotels, bookings made more than **180 days** in advance have a **64.2% failure rate**.

Conversely, booking made within 7 days of arrival have only a **12%** cancellation rate.



### 4.2 The Deposit Fallacy

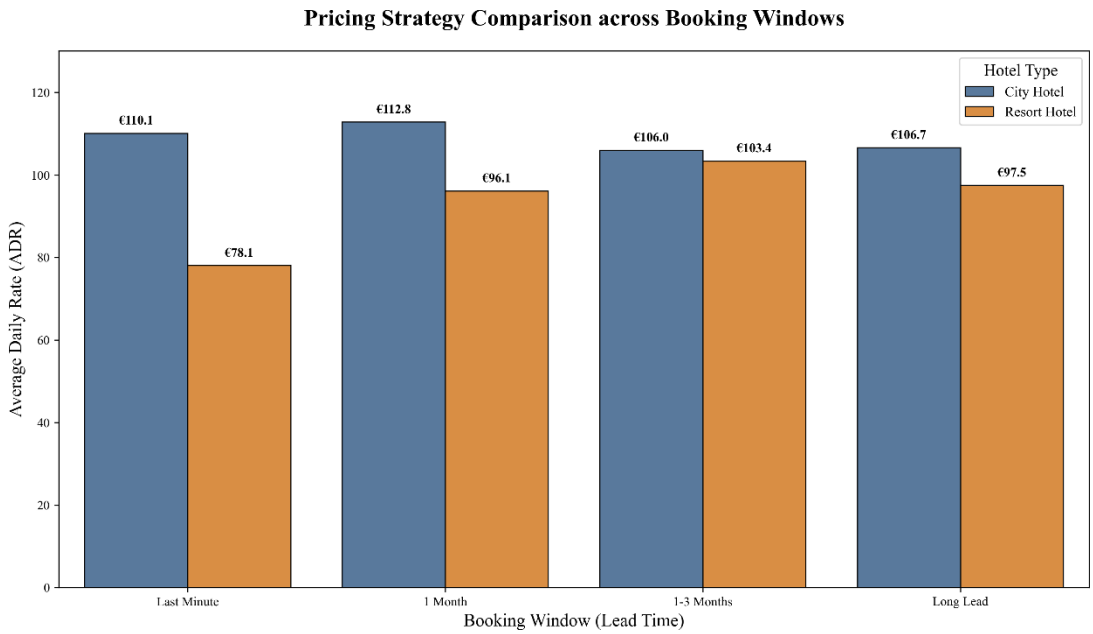
Contrary to intuition, **Non-Refundable** bookings in our data showed a **99.8% cancellation rate**. These are almost exclusively linked to "Group" segments and "Offline Travel Agents." This suggests a systemic failure in the B2B channel where entire blocks are reserved and subsequently defaulted on.

## 5. Phase 4: Revenue Mining (The Profit Engine)

### 5.1 Pricing Strategies: Harvest vs. Clearance

**City Hotels (Harvesting):** Effectively utilize a "Last-Minute Premium" strategy. ADR for bookings within 1 week is **€110**, compared to €106 for long-lead bookings.

**Resort Hotels (Clearance):** Suffer from "Perishable Inventory" panic. Last-minute ADR drops to **€78**, significantly lower than the €103 peak for 1–3-month bookings.




### 5.2 The "Family" Premium

**City Hotel**: Family ADR is **€149** vs. Adults-only €104.

**Resort Hotel:** Family ADR is **€154** vs. Adults-only €86.

# 6. Phase 5: Customer Personas & Loyalty

## 6.1 The Anchor of Stability: Repeated Guests

Repeated guests are the foundation of operational security. In Resort Hotels, their cancellation rate is a negligible **6.7%**, compared to 29% for new guests.

## 6.2 The "Contract" Segment Failure

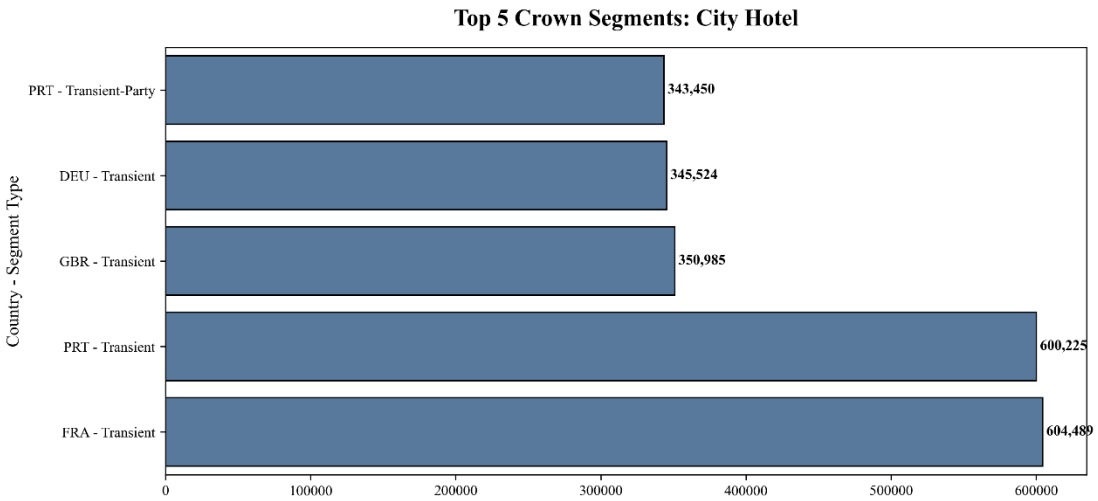
In City Hotels, **Contract** bookings (corporate agreements) are performing poorly with a **48% cancellation rate**, proving less reliable than standard transient (walk-in) customers.

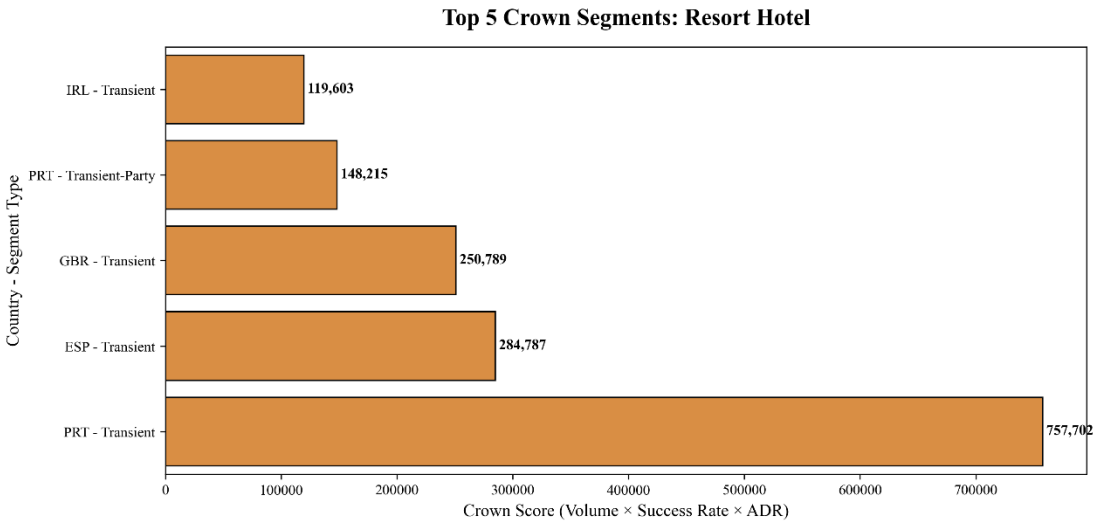
# 7. Phase 6: The Crown Jewel Analysis

To identify our most valuable markets, we developed a **Crown Score**:

$$Score = Total\ Volume\ \times Success\ Rate\ \times ADR$$

| Hotel | Country | Customer Type | Total Bookings | Success Rate | Average ADR | Crown Score |
|---|---|---|---|---|---|---|
| City Hotel | FRA | Transient | 6,978 | 75.95% | 114.05 | 604,489.01 |
| | PRT | Transient | 20,194 | 28.87% | 102.95 | 600,225.42 |
| | GBR | Transient | 4,636 | 66.98% | 113.04 | 350,985.65 |
| Resort Hotel | PRT | Transient | 13,698 | 57.02% | 97.02 | 757,702.72 |
| | ESP | Transient | 3,222 | 74.27% | 119.01 | 284,787.41 |
| | GBR | Transient | 3,841 | 77.84% | 83.88 | 250,789.04 |




# 8. Final Strategic Recommendations

## 8.1 City Hotel

**Tighten Cancellation Risk:** Implement mandatory non-refundable deposits for any booking with a Lead Time exceeding 90 days.

**Rationalize B2B Contracts:** Renegotiate terms with Offline Travel Agents and “Group” coordinators. The current 48%+ cancellation rate is unsustainable for inventory management.

**Capitalize on the “French Gold”:** Allocate a significant share of the digital marketing budget toward French transient travelers, who show the best balance of price and reliability.

## 8.2 Resort Hotel

**Implement Last-Minute Pricing:** Stop aggressive price slashing for late bookings. Instead, offer “Value-Add” (free spa, dinner) to maintain ADR while filling rooms.

**Cultivate the Family Premium Segment:** Since Families pay a €60+ premium, invest in “Kids-Club” amenities and family-centric marketing to increase the volume of this high-margin segment.

**The “British Anchor”:** Deepen relationships with GBR (UK) agents. Their 99.9% success rate in the Contract segment makes them the perfect “occupancy filler” for low seasons.